

```

In [1]: import numpy as np
import pandas as pd

# Create the Hotel dataset
data = {
    'CustomerID': [1, 2, 3, 4, 5, 6, 7, 8, 9, 9, 10],
    'Age_Group': ['20-25', '30-35', '25-30', '20-25', '35+',
                  '35+', '35+', '20-25', '25-30', '25-30', '30-35'],
    'Rating(1-5)': [4, 5, 6, -1, 3, 3, 4, 7, 2, 2, 5],
    'Hotel': ['Ibis', 'LemonTree', 'RedFox', 'LemonTree', 'Ibis',
              'Ibys', 'RedFox', 'LemonTree', 'Ibis', 'Ibis', 'RedFox'],
    'FoodPreference': ['veg', 'Non-Veg', 'Veg', 'Veg', 'Vegetarian',
                       'Non-Veg', 'Vegetarian', 'Veg', 'Non-Veg',
                       'Non-Veg', 'non-Veg'],
    'Bill': [1300, 2000, 1322, 1234, 989, 1909, 1000, 2999,
             3456, 3456, 6755],
    'NoOfPax': [2, 3, 2, 2, 2, 2, -1, -10, 3, 3, 4],
    'EstimatedSalary': [40000, 59000, 30000, 120000, 45000,
                        122220, 21122, 345673, -99999, -99999, 87777],
    'Age_Group.1': ['20-25', '30-35', '25-30', '20-25', '35+',
                    '35+', '35+', '20-25', '25-30', '25-30', '30-35']
}

df = pd.DataFrame(data)

# Display original dataset
print("Original Dataset:")
print(df)

# Check dataset information
print("\nDataset Information:")
df.info()

# Identify duplicate rows
print("\nDuplicate Rows:")
print(df.duplicated())

# Remove duplicate rows
df.drop_duplicates(inplace=True)

# Reset index
df.reset_index(drop=True, inplace=True)

print("\nDataset After Removing Duplicates:")
print(df)

# Remove unnecessary duplicate column
df.drop(columns=['Age_Group.1'], inplace=True)

# Replace inappropriate numerical values with NaN
df.loc[df['CustomerID'] < 0, 'CustomerID'] = np.nan
df.loc[df['Bill'] < 0, 'Bill'] = np.nan
df.loc[df['EstimatedSalary'] < 0, 'EstimatedSalary'] = np.nan

```

```

# Validate rating range: 1 to 5
df.loc[~df['Rating(1-5)'].between(1, 5), 'Rating(1-5)'] = np.nan

# Validate number of people
df.loc[~df['NoOfPax'].between(1, 20), 'NoOfPax'] = np.nan

# Correct hotel name
df['Hotel'] = df['Hotel'].replace({'Ibys': 'Ibis'})

# Standardize food preferences
df['FoodPreference'] = df['FoodPreference'].replace({
    'veg': 'Veg',
    'Vegetarian': 'Veg',
    'non-Veg': 'Non-Veg'
})

print("\nDataset After Correcting Inappropriate Data:")
print(df)

# Fill missing numerical values
df['CustomerID'] = df['CustomerID'].fillna(
    df['CustomerID'].median()
)
df['Bill'] = df['Bill'].fillna(round(df['Bill'].mean()))
df['EstimatedSalary'] = df['EstimatedSalary'].fillna(
    round(df['EstimatedSalary'].mean())
)
df['NoOfPax'] = df['NoOfPax'].fillna(
    round(df['NoOfPax'].median())
)
df['Rating(1-5)'] = df['Rating(1-5)'].fillna(
    round(df['Rating(1-5)'].median())
)

# Fill missing categorical values with mode
df['Age_Group'] = df['Age_Group'].fillna(
    df['Age_Group'].mode()[0]
)
df['Hotel'] = df['Hotel'].fillna(df['Hotel'].mode()[0])
df['FoodPreference'] = df['FoodPreference'].fillna(
    df['FoodPreference'].mode()[0]
)

# Display final dataset
print("\nFinal Cleaned Dataset:")
print(df)

# Check remaining missing values
print("\nMissing Values After Preprocessing:")
print(df.isnull().sum())

print("\nData Preprocessing Completed Successfully!")

```

Original Dataset:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	\
0	1	20-25	4	Ibis	veg	1300	
1	2	30-35	5	LemonTree	Non-Veg	2000	
2	3	25-30	6	RedFox	Veg	1322	
3	4	20-25	-1	LemonTree	Veg	1234	
4	5	35+	3	Ibis	Vegetarian	989	
5	6	35+	3	Ibys	Non-Veg	1909	
6	7	35+	4	RedFox	Vegetarian	1000	
7	8	20-25	7	LemonTree	Veg	2999	
8	9	25-30	2	Ibis	Non-Veg	3456	
9	9	25-30	2	Ibis	Non-Veg	3456	
10	10	30-35	5	RedFox	non-Veg	6755	

	NoOfPax	EstimatedSalary	Age_Group.1
0	2	40000	20-25
1	3	59000	30-35
2	2	30000	25-30
3	2	120000	20-25
4	2	45000	35+
5	2	122220	35+
6	-1	21122	35+
7	-10	345673	20-25
8	3	-99999	25-30
9	3	-99999	25-30
10	4	87777	30-35

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 11 entries, 0 to 10

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	CustomerID	11 non-null	int64
1	Age_Group	11 non-null	str
2	Rating(1-5)	11 non-null	int64
3	Hotel	11 non-null	str
4	FoodPreference	11 non-null	str
5	Bill	11 non-null	int64
6	NoOfPax	11 non-null	int64
7	EstimatedSalary	11 non-null	int64
8	Age_Group.1	11 non-null	str

dtypes: int64(5), str(4)

memory usage: 1.1 KB

Duplicate Rows:

0	False
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False
9	True

10 False
dtype: bool

Dataset After Removing Duplicates:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	\
0	1	20-25	4	Ibis	veg	1300		2
1	2	30-35	5	LemonTree	Non-Veg	2000		3
2	3	25-30	6	RedFox	Veg	1322		2
3	4	20-25	-1	LemonTree	Veg	1234		2
4	5	35+	3	Ibis	Vegetarian	989		2
5	6	35+	3	Ibys	Non-Veg	1909		2
6	7	35+	4	RedFox	Vegetarian	1000		-1
7	8	20-25	7	LemonTree	Veg	2999		-10
8	9	25-30	2	Ibis	Non-Veg	3456		3
9	10	30-35	5	RedFox	non-Veg	6755		4

	EstimatedSalary	Age_Group.1
0	40000	20-25
1	59000	30-35
2	30000	25-30
3	120000	20-25
4	45000	35+
5	122220	35+
6	21122	35+
7	345673	20-25
8	-99999	25-30
9	87777	30-35

Dataset After Correcting Inappropriate Data:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	\
0	1.0	20-25	4.0	Ibis	Veg	1300.0	
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	
2	3.0	25-30	NaN	RedFox	Veg	1322.0	
3	4.0	20-25	NaN	LemonTree	Veg	1234.0	
4	5.0	35+	3.0	Ibis	Veg	989.0	
5	6.0	35+	3.0	Ibis	Non-Veg	1909.0	
6	7.0	35+	4.0	RedFox	Veg	1000.0	
7	8.0	20-25	NaN	LemonTree	Veg	2999.0	
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	
9	10.0	30-35	5.0	RedFox	Non-Veg	6755.0	

	NoOfPax	EstimatedSalary
0	2.0	40000.0
1	3.0	59000.0
2	2.0	30000.0
3	2.0	120000.0
4	2.0	45000.0
5	2.0	122220.0
6	NaN	21122.0
7	NaN	345673.0
8	3.0	NaN
9	4.0	87777.0

Final Cleaned Dataset:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	\
0	1.0	20-25	4.0	Ibis	Veg	1300.0	

1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0
2	3.0	25-30	4.0	RedFox	Veg	1322.0
3	4.0	20-25	4.0	LemonTree	Veg	1234.0
4	5.0	35+	3.0	Ibis	Veg	989.0
5	6.0	35+	3.0	Ibis	Non-Veg	1909.0
6	7.0	35+	4.0	RedFox	Veg	1000.0
7	8.0	20-25	4.0	LemonTree	Veg	2999.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0
9	10.0	30-35	5.0	RedFox	Non-Veg	6755.0

	NoOfPax	EstimatedSalary
0	2.0	40000.0
1	3.0	59000.0
2	2.0	30000.0
3	2.0	120000.0
4	2.0	45000.0
5	2.0	122220.0
6	2.0	21122.0
7	2.0	345673.0
8	3.0	96755.0
9	4.0	87777.0

Missing Values After Preprocessing:

CustomerID	0
Age_Group	0
Rating(1-5)	0
Hotel	0
FoodPreference	0
Bill	0
NoOfPax	0
EstimatedSalary	0

dtype: int64

Data Preprocessing Completed Successfully!

In []: